CALCULATIONS:

TABLE 5C:
(AMPACITY CORRECTION FACTORS FOR TABLE 2)
CONDUCTORS = % CORRECTION FACTOR

TABLE 2:
(ALLOWABLE AMPACITIES FOR CONDUCTORS IN RACEWAY)

TABLE 16:
(MINIMUM SIZE CONDUCTORS FOR BONDING)
BONDING CONDUCTORS USED: COPPER WIRE, AWG

TABLE 6A:
(MAXIMUM CONDUCTORS IN CONDUIT)
CONDUCTORS USED: 600V RW90 XLPE – WITHOUT JACKET

LIGHTING PANEL 4A

TABLE 5C:
AMPACITY = A
(% CORRECTION FACTOR) = A

TABLE 2:
AMPACITY = A
CONDUCTOR SIZE = # AWG.

TABLE 16:
AMPACITY = A
BONDING CONDUCTOR SIZE = # AWG.

TABLE 6A:
CONDUCTORS, # AWG = mm EMT.

POWER PANEL 4B
PRIMARY (FEEDING TRANSFORMER):

TABLE 5C:
AMPACITY = A
(% CORRECTION FACTOR) = A

TABLE 2:
AMPACITY = A
CONDUCTOR SIZE = # AWG.

TABLE 16:
AMPACITY = A
BONDING CONDUCTOR SIZE = # AWG.

TABLE 6A:
CONDUCTORS, # AWG = mm EMT.

SECONDARY (FEEDING POWER PANEL):

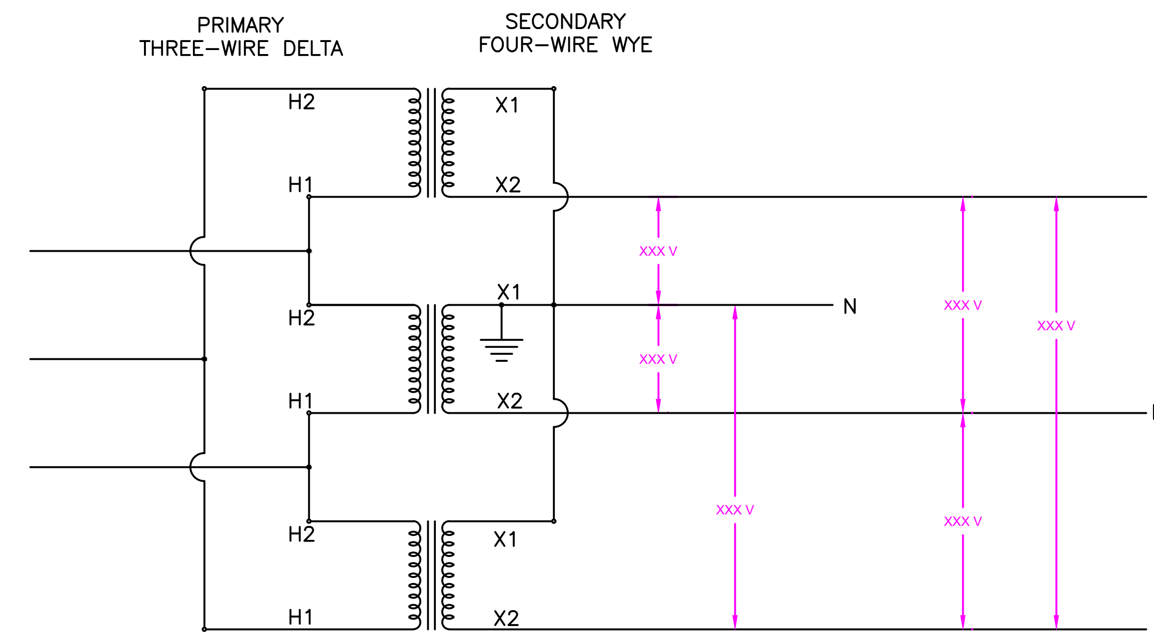
TABLE 5C:
AMPACITY = A
(% CORRECTION FACTOR) = A

TABLE 2:
AMPACITY = A
CONDUCTOR SIZE = KCMIL

TABLE 16:
AMPACITY = A
BONDING CONDUCTOR SIZE = # AWG.

TABLE 6A:
CONDUCTORS, KCMIL = mm EMT.

TRANSFORMER SIZE:
A (LOAD) * V = VA
(% POWER FACTOR) = VA
NEXT COMMON SIZE = KVA



DELTA–WYE TRANSFORMER
NOT TO SCALE

ALGONQUIN COLLEGE

ELECTRICAL DESIGN WITH AUTOCAD

COURSE NUMBER:	ELE8944
COURSE SECTION:	030
TITLE:	COMMERCIAL BUILDING ELECTRICAL DESIGN TRANSFORMERS CALCULATIONS
PROJECT:	PROJECT #2
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